Timur Khairulov

KhrTim | timurkhairulov@cau.ac.kr | (+82) 010-2181-6268
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LLM Engineer with hands-on experience in fine-tuning, optimization, and deployment of large language and vision-language models at research and prototype-production scale. Strong systems background in C++ and performance optimization, with research publications in deep learning and generative models.

Education

2024 - Present	MS in Artificial Intelligence at **Chung-Ang University** CAYSS scholarship recipient	(GPA: 4.22/4.5)
2018 - 2023	BS in Computer Science at **Electrotechnical University "LETI"**	(GPA: 4.3/5)
2022 - 2023	Exchange Semester at **Inha University**	(GPA: 3.9/4.5)

Selected LLM & AI Projects

VLM Compression: Optimization Techniques for Vision-Language Models GitHub | Blog
Python, PyTorch, BitsAndBytes | December 2025

- Benchmarked 8-bit and 4-bit quantization performance for BLIP-2 and LLaVA models
- Implemented structural pruning of MLP and KV attention components, reducing memory footprint and inference latency in vision-language models
- Uploaded pruned models to HuggingFace with comprehensive analysis published on blog

Curriculum Learning Fine-Tuning for Mathematical Reasoning GitHub | Blog
Python, PyTorch, Transformers, LoRA, HuggingFace | November 2025

- Fine-tuned SmolLM2 and Phi-3 models on GSM8K mathematical reasoning dataset using complexity-based curriculum learning strategies
- Achieved 1.37% accuracy improvement on difficult problems compared to baseline random-order training
- Designed custom curriculum scheduling within HuggingFace Trainer for reproducible fine-tuning
- Published comprehensive analysis with reproducible code and demo

BAGen: Background Animation Generation for Children's Songs GitHub
Python, PyTorch, Diffusion Models | January 2025

- Built end-to-end pipeline generating synchronized lyrics and animated backgrounds for children's songs using LLMs and diffusion models
- Developed production-ready system with GUI and CLI interfaces
- Submitted for publication in scientific reports

Information-Theoretic Feature Selection Library GitHub
MATLAB, Python | August 2024

- Implemented and configured 30+ unsupervised feature selection algorithms with 25 curated datasets
- Created extensible experimental framework for testing novel joint entropy maximization method
- Developed as foundation for Master's thesis research and future journal publication

Work Experience

C++ Software Engineer (Real-Time Systems), YADRO May 2023 - February 2024

- Developed real-time signal processing modules for 4G base stations using C++ and AVX SIMD instructions, improving throughput by 30% through optimized vectorization
- Optimized memory-intensive algorithms in telecommunications software, reducing latency by 25% through cache-aware programming and efficient data structure design
- Led development of L1-L2 communication visualization application with team of 3 developers, enabling project managers to analyze telecommunication delays using Python, TypeScript, and MongoDB
- Collaborated with cross-functional team of 8 engineers to deliver production-ready code meeting strict real-time performance requirements for deployed base stations
- Applied performance optimization principles later to ML workloads, including model inference latency and memory efficiency

Teaching Experience

Teaching Assistant, Computer Vision, Chung-Ang University Spring 2025

- Prepared course materials for lecture on CNN architectures and contributed to exam preparation and solution evaluation
- Assisted professor with grading and providing feedback on student assignments and exam solutions

Publications

Journal Articles

1. ChainImputer: A Neural Network-Based Iterative Imputation Method Using Cumulative Features
   MDPI Symmetry | 2025 | DOI: 10.3390/sym17060869
2. EP-REx: Evidence-Preserving Receptive-Field Expansion for Efficient Crack Segmentation
   MDPI Symmetry | 2025 | DOI: 10.3390/sym17101653

Conference Papers

1. Diffusion Model-Based Generative Pipeline for Children Song Video
   IEEE International Conference on Consumer Electronics (ICCE) | 2025 | IEEE Xplore
2. Review of current approaches in the area of object detection for autonomous vehicles
   Institute of Electronics and Information Engineers (IEIE) | 2024 | DBPIA

Technical Skills

Machine Learning	PyTorch, NumPy, Pandas, Scikit-learn, OpenCV, HuggingFace Transformers
Programming	C/C++, Python, MATLAB, SQL
Development	Git, Linux, CMake, Docker, Weights & Biases
Systems	Parallel Programming (AVX), Operating Systems, Computer Architecture
Languages	English (IH - OPIc Certificate), Russian (Native), Korean (Basic)

Awards and Honors

2024 - Present	CAYSS Scholarship Recipient at **Chung-Ang University**
2018 - 2023	Bachelor's Degree Scholarship Recipient at **Electrotechnical University "LETI"**